
Impact of a Four-Day Head-Down Tilt (-6°) on Lidocaine Pharmacokinetics Used as Probe to Evaluate Hepatic Blood Flow

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The impact of a microgravity simulation using a head-down tilt (-6°) on lidocaine pharmacokinetics used as a probe to evaluate hepatic blood flow is discussed. Eight healthy male subjects were selected for a 7-day study, including a 4-day head-down tilt from day 2 to day 5. Subjects were given 1 mg/kg of lidocaine on days 1 through 5 and 7. Blood sampling, cardiac output, and hepatic artery blood flow velocity measurements were done within 6 hours after administration. Cardiac output increased significantly during head-down tilt, and returned to basal values during the recovery period. Blood flow velocity in the hepatic artery increased during the first day of the down tilt. Slight side effects (buzzing noise in the ears and sleepy feeling) were reported within minutes after the injection of lidocaine. Lidocaine disposition was modified during head-down tilt: a significant decrease in maximal concentration (1.47 ± 0.26 mg/L on day 1 and 0.96 ± 0.30 mg/L on day 2); an increase in elimination clearance from 8.24 ± 3.22 mL/kg $\cdot$ minutes $^{-1}$ to 11.63 ± 3.00 mL/kg $\cdot$ minutes $^{-1}$; an increase in volume of distribution on day 2 and a decrease to lower than basal value on the other days (2.77 ± 1.73 L/kg on day 1 and 2.33 ± 0.48 L/kg on day 7). Half-life regularly decreased from 264 ± 210 minutes to 160 ± 60 minutes between day 1 and day 7. With the exception of day 1, no correlation was found between blood flow in the hepatic artery and lidocaine clearance, probably owing to the lack of precision of echo Doppler measurement, which only permitted indirect assessment. The reported pharmacokinetic modifications suggested that the dosage of drugs such lidocaine must be adapted during long-term administration in microgravity conditions by increasing the dose or reducing the interval of drug intake.

Microgravity is a different condition for human beings, which leads to many physiologic modifications. These are described in function of time: an acute phase during the first flight days followed by an adaptive phase, which may require days or weeks.¹

The cardiovascular modifications are among the most important during the first period. Because of the

loss of hydrostatic pressure, a fluid shift of extra- and intravascular volumes to the upper part of the body occurs. This increase in blood volume limited to the upper part of the body is considered as a hypervolemia by baroreceptors. In 24 to 48 hours, neurohormonal adaptation induces an actual hypovolemia. The fluid shift and the resulting cardiovascular adaptation syndrome could have profound consequences on drug disposition, as was previously shown in theoretical studies.²⁻³ For proper use of drugs during space flights, it appeared that it was necessary to investigate the different pharmacokinetic steps by using probe drugs during head-down tilt (-6°). This model is thought to well simulate physiologic changes, particularly cardiovascular adaptation syndrome occurring during microgravity.⁴⁻⁵

Lidocaine is an anti-arrhythmic drug, its elimination is directly correlated to hepatic blood flow.⁶⁻⁷

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Variations of this parameter are the limiting step for the elimination of drugs with a high hepatic extraction ratio.⁸ In this paper, the authors report the pharmacokinetic profiles of lidocaine before, during, and after a 4-day head-down tilt associated with assessments of cardiac output and blood flow velocity in the hepatic artery using the echo-Doppler technique.

SUBJECTS AND METHODS

Study Design

After approval by the local ethics committee, 8 healthy male volunteers aged between 25 to 45 years, non- or low smokers (less than 5 cigarettes per day) gave their informed consent and were included in the study. Characteristics of the subjects were the following: age, 35.5 ± 8.2 years; height, 174.1 ± 7.2 cm; weight, 72.4 ± 8 kg. No subject was allowed to smoke or take medication except lidocaine at the time of the study. The experiment was performed during 7 days: ambulatory on days 1 and 7, and in head-down tilt between day 2 and day 5. No administration was done on day 6; the subjects stood up in the morning and were free to walk or to sit until the evening. During the ambulatory phases, subjects could stretch out during a maximum of 90 minutes after the last blood sampling. Head-down tilt began on day 2 at 7:30 AM in strict head-down tilt at -6° , the position being monitored by video camera. The subjects were authorized to eat a breakfast without lipids and a fruit juice before the injection as well as 200 mL of water during the 4 hours after the injection. A standard meal was taken 4 hours after the injection.

Clinical Measurements

Every day, clinical symptoms were noted; weight and hydric balance (liquid intake and diuresis) were measured. During the injection of lidocaine and the following hour, the electrocardiogram and the blood pressure in the humeral artery (Dinamap; Critikon, Tampa, FL) were continuously monitored each minute during the first 15 minutes and then every 5 minutes.

The subjects stood up at day 6 in the morning after half an hour in head-down tilt (0°) and a phase in sitting posture. They were standing during 10 minutes while electrocardiogram (ECG) and blood pressure were monitored.

Cardiac Output Measurement

Cardiac output was measured by impedancemetry (cardiodynamic data processing system with pace-

mate module; BioMed Medical Manufacturing, Irvine, CA). This technique was described by Kubeck and then simplified by Sramek.⁹ In this study, the variations of the cardiac output indexed to corporal surface (cs) ($\text{l/min/m}^2\text{cs}$) were analyzed. Two series of measurements were performed: the first ones, in the morning before the injection in sitting position on day 1, day 2 (before the head-down tilt) and day 7, and in head-down tilt on days 3, 4, and 5; the second ones, about 3 hours after the injection, in sitting position on days 1 and 7, and in head-down tilt from day 2 to day 5.

Blood Flow Velocity in Hepatic Artery

Blood flow velocity was measured in the hepatic artery by echo Doppler technique.¹⁰⁻¹¹ At each occasion, the hepatic artery was first located by echography, and then the velocity was measured with a pulsed Doppler (Echo Doppler Ultra Mark 4; Advanced Technology Laboratories, Bothell, WA). The diameter of the artery was measured, and the mean velocity calculated from the spectral outline was analyzed. With the hypothesis of a constant diameter of the artery, the variations of the mean velocity were proportional to the variations of the blood flow, so this Doppler technique allowed an indirect assessment of the hepatic blood flow.¹²

Pharmacokinetic Measurements

On days 1 to 5 and 7, 1 mg/kg of body weight (minimal therapeutic dose of lidocaine) was administered in 3 minutes intravenously in the morning at 8.00 AM. Blood was sampled in heparinized tubes (2 mL), before and at 5, 15, 30, 45, 60, 90, 120, 180, 240, 300, and 360 minutes after the drug administration. Blood samples were immediately centrifuged and plasma was frozen at -20°C until assayed.

Plasma concentrations were measured using a homogenous enzyme immunoassay (Emit Lidocaine, Syva, Dardilly, France). Briefly, lidocaine competes with enzyme-linked lidocaine against monoclonal antibodies. The labeled unbound glucose-6-phosphate lidocaine reacts on glucose-6-phosphate and converts NAD into NADH, of which absorbance proportionally varies at 340 nm. The commercial kit was adapted and validated to measure lidocaine between 0.05 and 12 mg/L. The intraday precision and inaccuracy were 2.3% – 15.5%, and +0.5% – +3.1%, respectively. The interday precision and inaccuracy were 3.4% – 14.4% and -7.4% – $+7.2\%$, respectively. During the sample measurements, the quality control precision (3 levels) varied from 2.4% to 10.7%

and their inaccuracy from -1.4% and $+6.2\%$. The limit of quantification was 0.05 mg/L .

Pharmacokinetic calculations were done using the SIPHAR program, release 4.0b.¹³ An adjustment using the least square method was performed according to an open two-compartmental model followed by a nonlinear fitting using the Powell algorithm and the inverse of the concentration as weighting factor. The C_{max} was defined as the observed maximal concentration, AUC as the area under the time concentration curve calculated by the trapezoidal method and extrapolated to infinity by the ratio of the last concentration to the slope of the terminal phase; CL as the clearance calculated by the ratio of the administered dose to the AUC, and V_z as the volume of distribution calculated by the ratio of the clearance to the slope of the terminal phase.

Statistical Analysis

For all statistical calculations, the significant level was $P < .05$, and the results were expressed as mean $\pm$ SD. Clinical data, log-transformed C_{max} , log-transformed AUC clearance, and V_z were compared using a repeated two-way analysis of variance taking into account "day" and "subject" factors. Simple contrasts were used when a significant difference was observed for estimating day to day variations. For pharmacokinetic parameters, 90% confidence intervals were also calculated.

Relationships between physiologic measurements and pharmacokinetic parameters were explored using regression curves.

RESULTS

Clinical and Physiologic Results

During and in the minutes following the injection, 7 of the 8 subjects (a few times for 3 of them, and at each injection for 4 of them) presented side effects such as a buzzing noise in the ear, a sleepy feeling, and a dry mouth. These symptoms disappeared quickly within 1 to 3 minutes. No modification of the ECG was noticed.

During the standing up phase after 4 days of head-down tilt, 3 subjects presented a significant decrease in systolic blood pressure ($>25 \text{ mm Hg}$), at minute 3 for 2 of them and at minute 8 for the last one. None of them presented clinical symptoms due to the arterial hypotension.

Concerning hydric balance, changes in diuresis were slight. It increased the first day of head-down tilt, and then decreased in the recovery period (significant difference between day 2 and day 7) as

shown in Figure 1. Liquid intake varied significantly during the head-down tilt with a decrease during the head-down tilt and an increase during the recovery period as shown in Figure 1.

A decrease in the weight of the subjects was noticed during the head-down tilt (-0.83%) with a significant difference between day 1 ($72.96 \pm 8.38 \text{ kg}$) and day 2 ($72.92 \pm 8.40 \text{ kg}$) versus day 5 ($72.35 \pm 8.10 \text{ kg}$).

Cardiac Output Measurements

The cardiac output measurements are shown on Figure 2. From the control period ($2.47 \pm 0.33 \text{ L/min/m}^2\text{cs}$), it increased significantly during the whole head-down tilt, especially on day 3 ($3.20 \pm 0.48 \text{ L/min/m}^2\text{cs}$), and decreased during the recovery period ($2.74 \pm 0.39 \text{ L/min/m}^2\text{cs}$); significant differences were observed between day 3 versus day 7 and day 5 versus day 7.

Blood Flow Velocity in Hepatic Artery

With the technique used, no significant variation in the diameter of the hepatic artery was observed during the experiment. Figure 3 shows an increase in mean blood flow velocity during the head-down tilt, essentially on the first day (day 2), and a decrease after the subjects stood up; significant differences were observed between day 2 ($0.39 \pm 0.23 \text{ m/sec}$), versus day 3 ($0.22 \pm 0.11 \text{ m/sec}$) and day 7 ($0.19 \pm 0.06 \text{ m/sec}$).

Pharmacokinetic Results

Figure 4 shows the mean profile of lidocaine plasma concentrations and Figure 5 depicts the variations of the mean $\pm$ SD of some pharmacokinetic parameters estimated during the study.

The C_{max} varied between $1.47 \pm 0.26 \text{ mg/L}$ on day 1 and $0.96 \pm 0.30 \text{ mg/L}$ on day 2, and then stabilized until the last day of the study. The statistical analysis showed a significant difference between day 1 and the other days of the experiment. The confidence interval approach showed that C_{max} did not differ by more than 20% for days 2, 4, 5, and 7. The AUC decreased from $130.69 \pm 47.65 \text{ mg/L} \cdot \text{minutes}^{-1}$ of day 1 to $92.51 \pm 21.43 \text{ mg/L} \cdot \text{minutes}^{-1}$ on day 5 with the same statistical results as those of C_{max} . Therefore, the elimination clearance increased from $8.24 \pm 3.22 \text{ mL/kg} \cdot \text{minute}^{-1}$ to $11.63 \pm 3.00 \text{ mL/kg} \cdot \text{minute}^{-1}$ between day 1 and day 2 and then stabilized at roughly $10 \text{ mL/kg} \cdot \text{minute}^{-1}$. Simple contrasts showed significant differences between day 1 and days 2, 5, and 7, and between day 2 and day 3. Clearances were

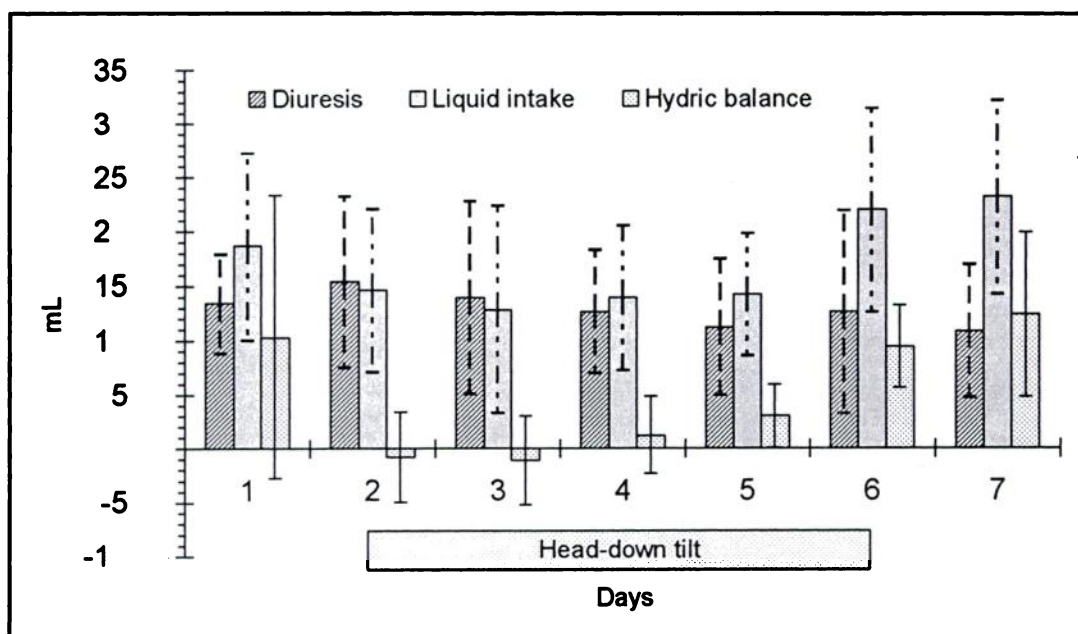


Figure 1. Diuresis, liquid intake, and hydric balance (mean $\pm$ SD) during the 7-day experiment.

equivalent on days 2, 5, and 7. The volume of distribution increased from 2.77 ± 1.73 L/kg on day 1 to 3.73 ± 2.52 L/kg on day 2, and returned near the basal value from day 3 to day 7 (2.33 ± 0.48 L/kg). Simple contrasts showed significant differences between day 2 and day 4, 5, and 7. Half-life regularly decreased from 264 ± 210 minutes to 160 ± 60 minutes between day 1 and day 7. The large variability observed on the first days of the experiment did not allow the demonstration of any difference over the experimental period for this parameter.

DISCUSSION

The observed clinical effects (buzzing noise in the ears and sleepy feeling) are those classically de-

scribed for this drug. In this study, no significant difference was shown between the maximal concentrations of the subjects with and without systemic clinical symptoms.

Three of the eight subjects presented an orthostatic hypotension, and all of them presented a large heart rate increase when they stood up. The increase in the heart rate associated with a drop in the blood pressure (particularly in systolic blood pressure) during the orthostatism were observed after the space flights and the simulation experiments.^{5,14-16} Like weightlessness, the head-down tilt induces a fluid shift from the lower extremities to the upper body part, which in turn provokes an increase in the

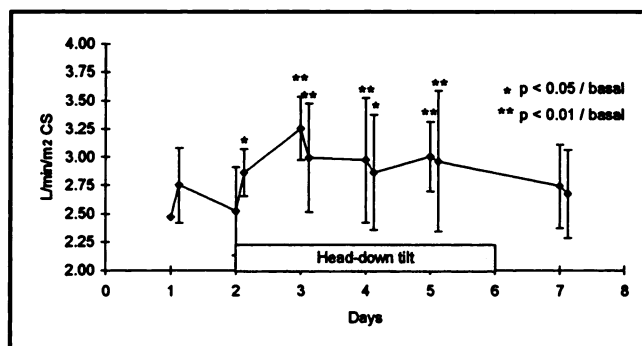


Figure 2. Cardiac output (mean $\pm$ SD) during the 7-day experiment.

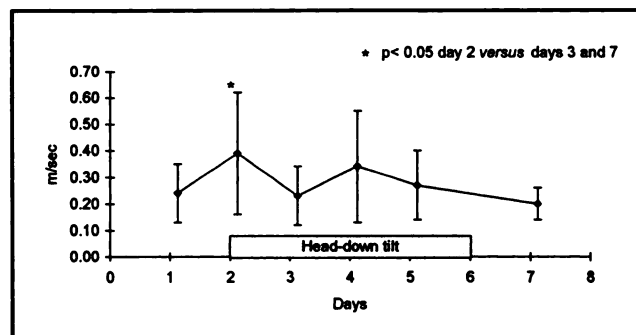


Figure 3. Hepatic blood flow velocity (mean $\pm$ SD) during the 7-day experiment.

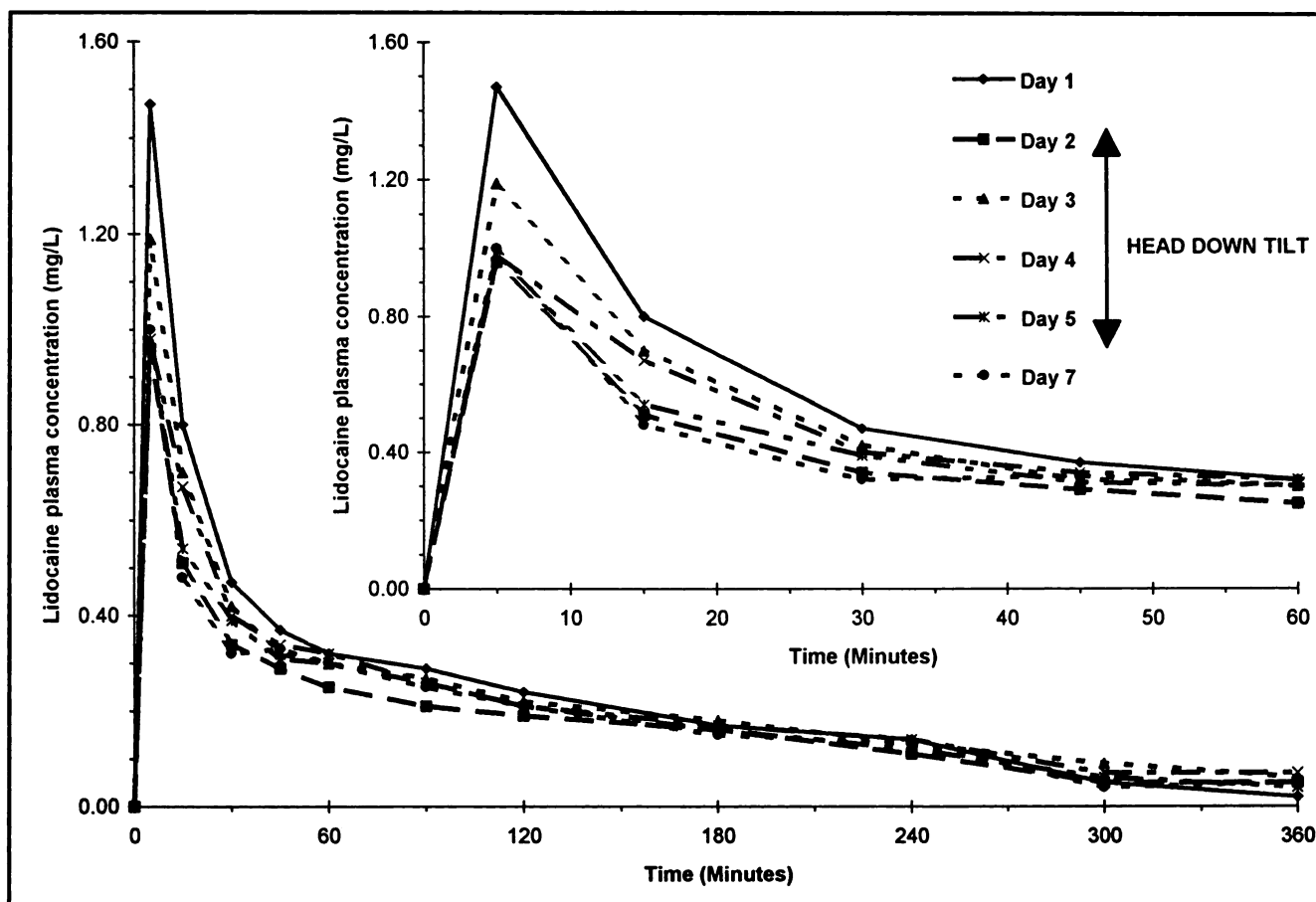


Figure 4. Mean profiles of lidocaine plasma concentrations.

secretion of atrial natriuretic peptide and a decrease in the secretion of the renin-angiotensin-aldosterone system, the antidiuretic hormone, and the catecholamines.¹⁹ This results in a marked and progressive loss of plasma volume.¹⁷⁻²¹ This hypovolemia is one of the factors of the orthostatic hypotension following space flights and weightless simulation experiments. Nevertheless, the assessment of the orthostatic tolerance was not the aim of this experiment, and the subjects were sitting before they stood up. This should have decreased the extent of orthostatic hypotension.

As observed in previous studies,²² blood pressure measured each day before lidocaine injection was decreased on day 2; this reduction was significant for diastolic measurements. That corresponds to the adaptation mechanisms during the first hours of head-down tilt with a probable decrease of the systemic vascular resistance in relation to the cardiovascular adaptation owing to the initial fluid

shift, as mentioned previously. The increase in the cardiac output during the head-down tilt (particularly on day 3) 24 hours after the head-down tilt began correlates with the increase in the cardiac output observed at the beginning of space flights and simulation experiments with ultrasonographic techniques.²³⁻²⁴ It was only after a few days that the cardiac output seemed to stabilize at a lesser level than in the preflight period.

The increase in diuresis and the decrease in the liquid intake with negative hydric balance observed on the first and the second days of the head-down tilt are in relation to the fluid shift and the neurohormonal mechanisms already described. The decrease in the diuresis and the increase in the liquid intake observed during the first day(s) of recovery relate to the hypovolemia. In the subjects, plasma volume tends to be restored. The decrease in weight noticed during the head-down tilt is consistent with the results of previous experiments where a slight but significant decrease in weight (−3%)

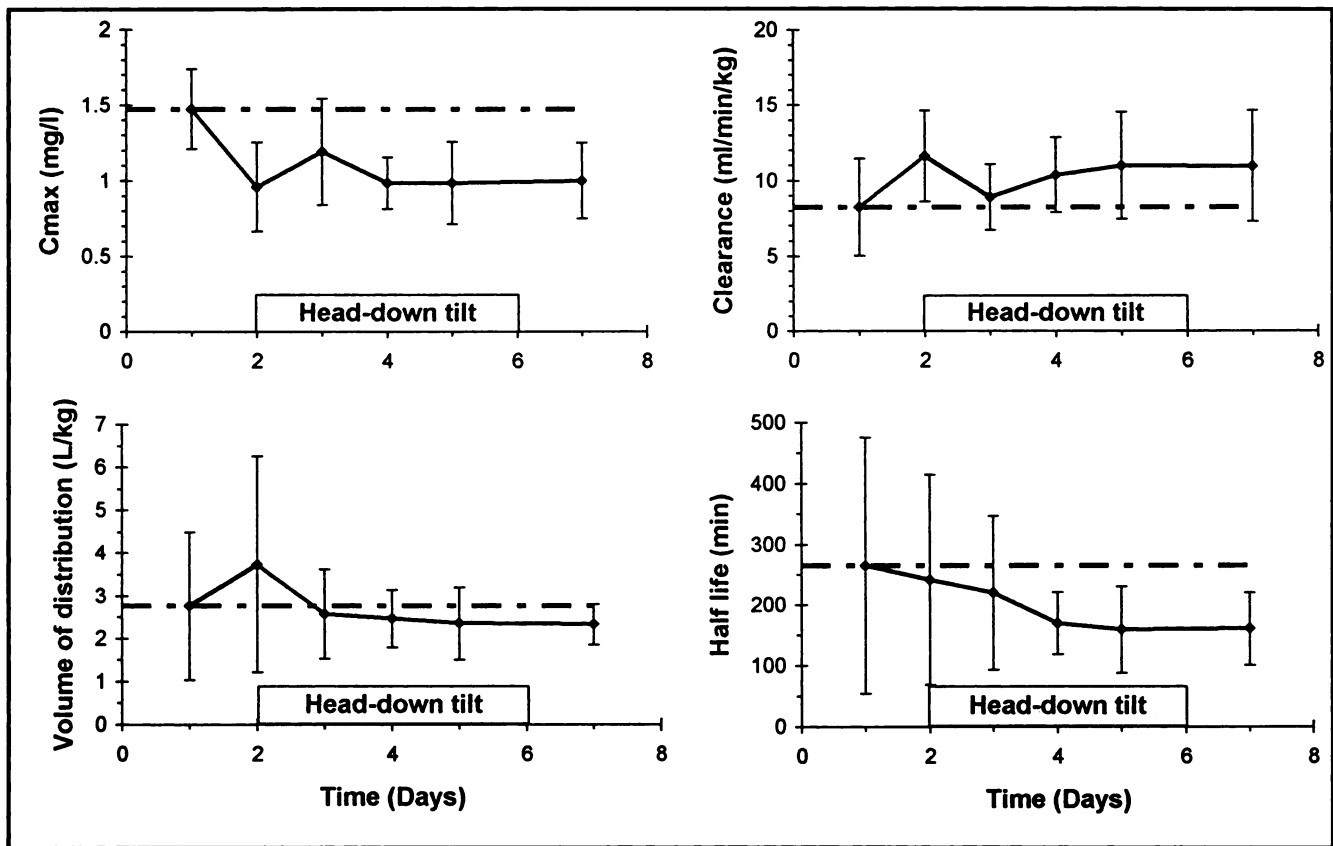


Figure 5. Pharmacokinetic parameters (mean $\pm$ SD) during the 7-day experiment.

was observed after 4 days of head-down tilt¹⁸⁻²⁰ essentially, in relation with loss of liquids.

The increase in blood flow velocities in the hepatic artery during the first day of head-down tilt (although not statistically significant) could be related to the fluid shift from the lower extremities towards the upper part of the body. The following nonsignificant changes are probably related to the adaptation of local vascular resistances. Several factors can modify the blood flow velocity in the hepatic artery by changing vascular resistance in the hepatic artery and portal vein area: posture, physical exercise, and food. In the experiment, the main factor was the posture change between the ambulatory phase and the head-down tilt. The subjects did not practice any muscular exercise and their diet did not change during the head-down tilt. Nevertheless, ultrasonographic techniques permitted blood flow velocity measurements and only indirect evaluation of blood flow. Changes in blood flow velocities are only proportional to blood flow changes if the artery diameter remains constant. But, the measurement of the ar-

tery diameter was not very precise, and some slight changes could have been underrated, making the results difficult to interpret.

The correlation between the cardiac output and the blood flow velocities in the hepatic artery during the head-down tilt was partial and varied along the head-down tilt period. This probably assessed different local vascular adaptations. The vascular adaptation in the hepatic area in microgravity was not studied. Some changes in blood flow velocities and vascular resistances were observed in other vascular areas (femoral artery).^{23,25}

Pharmacokinetically, the reported results on day 1 were comparable with those described in the literature.^{8,26} Lidocaine is well known and is used as a probe drug to investigate hepatic blood flow, because its clearance is directly related to this physiologic parameter.²⁶ In this study, no correlation was found between the elimination clearance and the hepatic artery flow rate with the exception of day 1 ($R = .6323$; $P = .06$). This lack of correlation could be owing to the inaccuracy of the ultrasonic technique. The ob-

served discordance on day 7 between the conserved elimination clearance and the decreased hepatic artery flow rate may be owing to a decrease in local hepatic vascular resistances to conserve a satisfactory hepatic blood flow. The lidocaine clearance variations observed in this study were similar to those reported by Feely et al.⁷ These authors showed that after the intravenous administration of a comparable dose of lidocaine (61 ± 4 mg) to idiopathic hypotensive patients, its clearance was reduced from 602 ± 101 mL/min to 475 ± 109 mL/min, respectively, when the drug was administered in standing posture compared with supine posture. They hypothesized that the hypotension could decrease the hepatic flow rate. Indocyanin green is also well known to be eliminated only by the liver and its clearance to be closely related to the hepatic blood flow. Putcha et al.²⁷ did not observe any variation in indocyanine green clearance during an experiment performed after a 24-hour head-down tilt. This result is not in opposition with the finding of this study, because on day 3, i.e., 24 hours after the beginning of the head-down tilt, no variation could be observed in the lidocaine clearance. Kates et al.²⁸ studied lidocaine at 100 mg by intravenous route in 12 subjects before and after 8 days of recumbency. They did not observe any modifications in lidocaine disposition and concluded that drug distribution and elimination are not changed in this situation.²⁸ It is well known that anti-orthostatic bedrest simulates cardiovascular modifications observed during weightlessness and induces higher variations than 0° recumbency.²⁹ This could explain the observed differences between this study and the present results.

A 35% increase in lidocaine volume of distribution was measured between the ambulatory period and the first head-down tilt value. It was followed by a 16% decrease on the last day of head-down tilt when compared with the ambulatory initial value. This observation may be partially explained by plasma volume modifications. In fact, the volume of distribution of an intravenous administered drug can be estimated as the sum of the plasma volume and tissue volumes weighed by the ratio of the free fraction of the drug in plasma to those in tissues. Different authors have shown an increase in plasma volume during the first hours of anti-orthostatic head-down tilt experiment owing to interstitial liquids migration.¹⁸⁻²⁰ This hypervolemia leads to a reactional hypovolemia, and after an anti-orthostatic 28-day head-down tilt plasma volume decreased by 11%.³⁰ On day 7, the persistence of the volume of distribution reduction was related to the uncompleted plasma volume reconstitution. Feely et al.⁷ have also reported an increase in lidocaine volume of distribution when measured in supine position, which cannot be explained by a variation

of drug binding to plasma proteins. The plasma volume reduction is unlikely to be the only explanation of the volume of distribution decrease, because the plasma volume only represents 3% of the lidocaine volume of distribution. In fact, lidocaine is bound to albumin (30%) and even more to α -1-glycoprotein (70%).²⁶ Therefore it should be interesting to investigate the variations of the concentrations of these two proteins during weightlessness and their consequences on lidocaine binding.

Half-life is proportional to the volume of distribution and inversely proportional to the clearance; therefore the observed decrease in lidocaine half-life was due to both the increase in its clearance and the reduction of its volume of distribution. At the end of the experiment, half-life was decreased by 40% when compared with the ambulatory period.

Space flights are increasing in number and duration so that a great use of drugs is more likely to occur. Thus, the knowledge of drug disposition during microgravity has to be known for their proper use during emergency or prolonged administration. According to the hypothesis reported in the authors' review about the influence of weightlessness on drug pharmacokinetics,²⁻³ they have observed variations in the disposition of lidocaine during a simulation of microgravity conditions using anti-orthostatic head-down tilt as a model. This drug is characterized by a low plasma protein binding, a large volume of distribution, and a clearance of elimination related to the hepatic blood flow. For this type of drug, such pharmacokinetic results may have consequences on their therapeutic efficacy. During an acute administration, the observed reduction of C_{max} may require an increase in administered dose to obtained plasma concentration in the therapeutic range of 2 to 5 mg/L and lower than the toxic concentration of 8 mg/L.²⁶ During prolonged administration, an adaptation of the dosage is likely to be necessary by increasing the dose and/or reducing the interval of drug intake in microgravity conditions.

Drugs such as alprenolol, desimipramine, meperidine, nifedipine, and propranolol show the same pharmacokinetic characteristics. They have already been on board during space flights or could be used in the future; they should then have specific regimens. Other studies must be planned to investigate different kinds of pharmacokinetic characteristics and then to predict their use in space, warranting the best safety and efficacy as possible.

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